The minimum number of edges in a pancyclic graph and the number of cycles in a graph

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Abstract

A graph on n vertices is pancyclic if it has a cycle of every length from 3 to n. Let $n + h(n)$ be the least number of edges of such a graph. A recent theorem of KNT states that $h(n) = \log_2 n + \log^* n + O(1)$, which settles Erdős Problem #1016. The proof rests on one estimate. Remove a small union of cycles from a connected graph of large cyclomatic number. Then a uniformly random even subgraph, restricted to the remaining edges, is a forest with probability at most $\frac{1}{2} + o(1)$.

We show that the same estimate bounds the number of cycles: a graph of cyclomatic number r has at most $(\frac{1}{2} + o(1))\, 2^r$ cycles. This answers, asymptotically, a question of Entringer and Slater from 1981. The best earlier bound for cubic graphs was $\frac{15}{16} 2^r$. Our bound is asymptotic only. It is below the trivial bound 2^r only when $\log_2 \log_2 \log_2 r > 42$.

We also give a short account of the pancyclic theorem, and some elementary results on the regime in which the underlying count is exact. The pancyclic theorem, the estimate and the bound on the number of cycles are formalised in Lean 4. The elementary results are not.

1 Introduction

1.1 Pancyclic graphs

A graph on n vertices is *pancyclic* if it contains a cycle of every length $3, 4, \dots, n$. Let $n + h(n)$ be the least number of edges of a pancyclic graph on n vertices. In this paper log is the binary logarithm, $\log^{(j)}$ is its j-fold iterate, and $\log^* x$ is the least $j \geq 0$ with $\log^{(j)} x \leq 1$. Graphs are finite and simple unless we say otherwise.

Theorem 1.1 (KNT [10])**.** *There is an absolute constant A such that, for every $n \geq 3$,*

$$\log n + \log^* n - A \;\leq\; h(n) \;\leq\; \lfloor \log n \rfloor + \log^* n + 1.$$

Bondy [5] stated without proof that $\log(n-1) - 1 \leq h(n) \leq \log n + \log^* n + O(1)$. His lower bound is a count. A connected graph with $n + h$ edges has at most $2^{h+1} - 1$ cycles, and a pancyclic graph needs $n - 2$ of them. Griffin [8] wrote this count out and determined $h(n)$ for $n \leq 37$. Problem #1016 in Bloom's collection of Erdős problems [4] asks whether the upper bound is the truth, and whether even $h(n) - \log n \to \infty$. Theorem 1.1 answers both questions. The paper [10] gives more of the history.

*Astra and Fable 5.2 are language models: GPT-6 Astra and Claude Fable 5.2. Section 1.5 says who did what, and which statements a machine has checked.

1.2 The cycle space and the forest estimate

For a graph G, an *even subgraph* is a set of edges that meets every vertex an even number of times. With symmetric difference as addition, the even subgraphs form a vector space $\mathcal{Z}(G)$ over $\mathbb{F}_2$, the *cycle space*. Its dimension is

$$\beta(G) = |E(G)| - |V(G)| + \mathrm{comp}(G),$$

where $\mathrm{comp}(G)$ is the number of components. We call $\beta(G)$ the *rank* of G. Other names are cyclomatic number and cycle rank. A *cycle* is a connected 2-regular subgraph. We identify a subgraph that has no isolated vertices with its edge set. Then a cycle is a nonzero element of $\mathcal{Z}(G)$, so a graph of rank r has at most $2^r - 1$ cycles.

Bondy's count uses nothing else. To go further, fix a subgraph W of G. For an even subgraph ξ of G we call $\xi \setminus E(W)$ its *outside part*. If a cycle uses an edge of W, then its outside part is a proper subset of a cycle, so it is a forest. The estimate behind Theorem 1.1 says that at most about half of all outside parts are forests.

Theorem 1.2 (Forest estimate, KNT [10])**.** *There is r_0 such that the following holds for all $r \geq r_0$. Let G be a connected simple graph of rank r. Let $W \subseteq G$ be a nonempty union of cycles with at most $\sqrt{r}$ edges and at most $(\log r)/100$ components. If ξ is a uniformly random element of $\mathcal{Z}(G)$, then*

$$\mathbb{P}(\xi \setminus E(W) \text{ is a forest}) \;\leq\; \frac{1}{2} + \frac{20}{\log^{(3)} r}. \tag{1}$$

The value $\frac{1}{2}$ has a simple source. Suppose that W meets the rest of G in two vertices s and t only. Then an outside part has even degree at every vertex, or it has odd degree exactly at s and t. There are equally many outside parts of each kind, and one of the first kind is not a forest unless it is empty (Example 2.2). Theorem 1.2 says that no other way to attach W leaves a larger proportion of forests, up to the error term.

Here is how the estimate enters the proof of Theorem 1.1. Sort the cycles of G by their outside parts. Two cycles with the same outside part differ only inside W, so the cycles with a given outside part have at most $|E(W)|$ different lengths. If W has rank t, there are 2^{r-t} outside parts. So G has cycles of at most $|E(W)|\,2^{r-t}$ lengths, and Theorem 1.2 improves this by a factor of two when r is large. Section 4 applies this at ranks that grow like a tower of exponentials. Each level halves the number of lengths that a given rank can supply, so each level costs one more edge. This gives the term $\log^* n$. Section 5 gives the construction for the upper bound, and Section 6 proves Theorem 1.2.

1.3 The number of cycles

Let $\psi(r)$ be the largest number of cycles in a graph of rank r. We have seen that $\psi(r) \leq 2^r - 1$. Entringer and Slater [6] showed that the maximum is attained by cubic graphs. They found graphs with slightly more than 2^{r-1} cycles, and they asked whether this is essentially the maximum. Later papers quote this as the conjecture $\psi(r) \sim 2^{r-1}$; see for instance [9]. Aldred and Thomassen [2] proved that $\psi(r) \leq \frac{15}{16}2^r$, and that a planar graph of rank r has at most $2^{r-1} + o(2^{r-1})$ cycles. Alahmadi, Aldred, dela Cruz, Solé and Thomassen [1] proved that a graph with p vertices and $q > 2p + 3\log(3p)$ edges has at most 2^{q-p} cycles. For a connected graph, $2^{q-p} = 2^{r-1}$.

This left the sparse graphs, and in particular the cubic ones. For those, $\frac{15}{16}2^r$ was the best bound when [9] was written, and we have found nothing newer.

Theorem 1.3. *There is r_1 such that every graph of rank $r \geq r_1$ has at most*

$$\Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big) 2^r$$

cycles. Consequently $\psi(r) \sim 2^{r-1}$.

This is an asymptotic statement and nothing more. The bound is below the trivial bound 2^r only if $\log^{(3)} r > 42$, that is, if $r > 2^{2^{42}}$. It is below $\frac{15}{16}2^r$ only if $\log^{(3)} r > 48$. The thresholds r_0 and r_1 are not explicit. So for every rank that one can write down, $\frac{15}{16}2^r$ is still the best bound known. The theorem also says nothing about the second-order term. Horak and Kim [9] proved that $\psi(r) \geq 2^{r-1} + c\, r\, 2^{r/2}$ for some $c > 0$ and all large r.

The proof takes one page (Section 3). Choose about $\log\log r$ edges of G so that only a small part of the cycle space avoids all of them. Every other cycle uses a chosen edge. We want to apply Theorem 1.2 to a small union of cycles W that contains the chosen edges, but G need not have short cycles through them. So we add chords to G until every chosen edge lies on a triangle, and we let W be the union of these triangles. A chord doubles the cycle space. It also doubles the set that we count, so nothing is lost.

1.4 Exact counting

Theorem 1.2 forces a loss of one half when the rank is large. Our last result describes the opposite regime, in which nothing is lost.

Theorem 1.4. (a) *Let N be a connected graph and let $s \neq t$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from s to t. If it has exactly $2^{\beta(N)}$, then each of these paths has at least $\beta(N)$ edges, and deleting the edges of any one of them leaves a forest.*

(b) *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. Suppose that every outside part is the outside part of some cycle of G. Then $G - E(W)$ has no two edge-disjoint cycles, and $\beta(G) \leq \beta(W) + |V(W)| + 3$.*

Part (b) concerns the bound of $|E(W)|\, 2^{r-t}$ cycle lengths from above. It can be exact only while the rank of G exceeds the rank of W by at most $|V(W)| + 3$. Part (a) explains the shape of the construction in Section 5, and with it the term $\log^* n$ (Section 7). That section also proves a stability statement (Proposition 7.4).

1.5 Who did what, and what has been checked

KNT is a person. Astra and Fable 5.2 are language models. Theorems 1.1 and 1.2, their proofs, and the Lean formalisation of these proofs come from [10]. KNT obtained them in September 2026 with Astra, and has described that process in public [11]. In November 2026 KNT asked Fable 5.2 to look for a simpler proof. That search did not find a new proof of Theorem 1.2. It found Theorem 1.3, the results of Section 7, and the changes of presentation listed below. Fable 5.2 then wrote the Lean proof of Theorem 1.3 and drafted this paper. KNT directed the work at each step.

Papers written with language models have earned some distrust. The usual complaints are these. The proofs read well but hide gaps. The formal statement, when there is one, is not the statement in the paper. References are invented or unread. Small results are oversold. Nobody says what was checked, or by whom. We answer these points one by one.

Statement	Source	Lean declaration (namespace `Erdos1016`)
Theorem 1.1	[10]	`mainTheorem`; the explicit upper bound is `Problem1016.upper_bound_integer`
Theorem 1.2	[10]	`ShortProof.fewComponentForestEstimate`
Theorem 1.3, upper bound	this paper	`cycleCountTheorem`, `cycleCountTheorem_general`
Theorem 1.3, $\psi(r) \sim 2^{r-1}$	this paper	`cycleCountAsymptotic`
Theorem 1.4 and Section 7	this paper	none

Table 1: The main statements and their formal counterparts at commit `756d5cc` of the repository [12].

What a machine has checked. Table 1 lists the formal counterparts of the main statements. The Lean kernel accepts their proofs with the axioms `propext`, `Classical.choice` and `Quot.sound` only. This certifies the formal statements. It does not certify the prose in this paper. Section 3 follows the formal proof of Theorem 1.3 step by step. Sections 4, 5 and 6 follow [10], and the Lean code follows [10] as well. Our text differs from [10] in three places. One counting lemma (Lemma 2.1) serves both theorems. The link formula (Lemma 6.2) is proved by a count of dimensions, without contractions. The two facts about the non-backtracking matrix (Lemma 6.7) are derived from its definition.

Formal statements. Appendix A prints the Lean statements behind Theorem 1.3, with the definitions that they use, so that the reader can compare them with the text. Graphs and cycles are those of Mathlib. The thresholds are existentially quantified in Lean, as they are here. The appendix also gives the commands that repeat the check.

What no machine has checked. Theorem 1.4 and the rest of Section 7 have ordinary proofs only. These proofs are short, and they use nothing beyond linear algebra over $\mathbb{F}_2$ and one lemma of Erdős and Pósa, which we prove. We also tested these statements on small graphs (Appendix B). Such tests are sanity checks, not proofs. No referee has read this draft.

References. We compared the bibliographic data of every reference with the publisher's record, with Crossref or with arXiv. The one exception is [6], where we relied on the citations in [1] and [9]. We read [1], [8], [10], and the parts of [9] that concern cycles. Of [2] we read the abstract. We have not seen [3], [5], [6], [7] or [13]. What we say about [5] and [6] is taken from [1], [4], [9] and [10]. No proof in this paper depends on a source that we have not seen: the lemma from [7] is proved in Section 7, and the facts related to [3, 13] are proved in Section 6.

Novelty and size of the results. Theorem 1.3 is a corollary of Theorem 1.2, and the new part of its proof is one page. We looked for an earlier proof of $\psi(r) \sim 2^{r-1}$, and for any bound below $\frac{15}{16}2^r$ for cubic graphs. We used web searches and the literature discussed in [1] and [9], and we found neither. A search of this kind can miss things. The results of Section 7 are elementary. One of them is known in the language of codes (see the remark after Proposition 7.2), and others may be folklore.

2 Outside parts

Let W be a subgraph of a graph G. Put $r = \beta(G)$, $t = \beta(W)$ and $m = |E(W)|$. The map $\xi \mapsto \xi \setminus E(W)$ is linear on $\mathcal{Z}(G)$, and its kernel is $\mathcal{Z}(W)$. So there are 2^{r-t} outside parts, and each of them is the outside part of exactly 2^t even subgraphs. If ξ is uniform on $\mathcal{Z}(G)$, then its outside

part is uniform on the set of outside parts. Let

$$p(G, W) = \mathbb{P}\big(\xi \setminus E(W) \text{ is a forest}\big).$$

This is the proportion of outside parts that are forests. Exactly $p(G, W)\, 2^r$ even subgraphs of G have a forest as outside part.

Lemma 2.1 (Counting lemma)**.** *Let $p = p(G, W)$.*

(a) *If a cycle C of G contains an edge of W, then $C \setminus E(W)$ is a forest. Hence at most $p\, 2^r$ cycles of G contain an edge of W.*

(b) *The cycles of G have at most $2^{r-t}(pm + 1 - p)$ different lengths.*

Proof. (a) A proper subset of the edge set of a cycle contains no cycle. The second statement follows, because distinct cycles are distinct even subgraphs.

(b) Let x be an outside part. Suppose first that x is not a forest, and let C be a cycle with outside part x. Then x contains a cycle and $x \subseteq C$, which forces $C = x$. So x accounts for at most one cycle, and hence for at most one length. Now suppose that x is a forest, and let C be a cycle with outside part x. Then C contains an edge of W, because C is not a forest. The length of C is $|x| + |C \cap E(W)|$, and $1 \le |C \cap E(W)| \le m$. So x accounts for at most m lengths. There are $p\, 2^{r-t}$ outside parts that are forests and $(1 - p)\, 2^{r-t}$ that are not. □

Example 2.2 (Two points of attachment)**.** Suppose that $G = W \cup N$, where W and N are connected, have no common edge, and have exactly two common vertices s and t. An $\{s, t\}$-*join* of N is a set of edges of N in which s and t have odd degree and all other vertices have even degree. The outside parts are the $2^{\beta(N)}$ even subgraphs of N and the $2^{\beta(N)}$ joins of N. A nonempty even subgraph is not a forest. Hence

$$p(G, W) \;\le\; \tfrac{1}{2} + 2^{-(r-t)},$$

with equality if and only if every join of N is a forest. So (1) holds without an error term when W is attached at two vertices.

For a concrete case, let N be a chain of k four-cycles, each joined to the next at opposite vertices, and let s and t be the two ends of the chain (Figure 1). Every join of N is a path from s to t, so equality holds. If W is a triangle and k is large, this graph satisfies the hypotheses of Theorem 1.2. So the constant $\frac{1}{2}$ in (1) cannot be lowered.

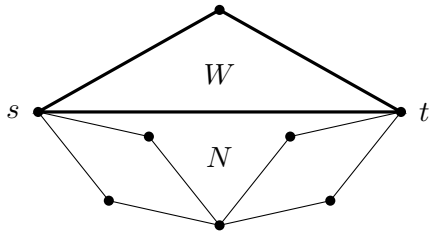


Figure 1: Example 2.2 with $k = 2$ and a triangle W (thick). Here $r = 4$ and $t = 1$. The eight outside parts are the four even subgraphs of N and the four paths from s to t in N. Five of them are forests, so $p(G, W) = 5/8$. Of the 16 even subgraphs of G, ten have a forest as outside part: the nine cycles that use an edge of W, and the empty set.

3 The number of cycles

This section proves Theorem 1.3. Appendix A names the Lean declaration for each step.

3.1 The lower bound

Let $k \geq 2$. Take a cycle of length k, and replace each of its edges by a four-cycle in which the two ends of the edge are opposite vertices. The new graph has $3k$ vertices and $4k$ edges, so its rank is $k+1$. Its cycles are the k four-cycles and the 2^k cycles that use one half of each four-cycle. So $\psi(r) \geq 2^{r-1} + r - 1$ for $r \geq 3$. Examples with slightly more than 2^{r-1} cycles go back to [6], and the bound of Horak and Kim in the introduction is stronger. The formal proof uses a closed chain of shortcuts from the construction of Section 5 instead. It has rank $j+1$ and at least 2^j cycles.

3.2 The upper bound

In this subsection G is a connected simple graph of rank r, and t is an integer with $1 \leq t \leq r$.

Lemma 3.1 (Independent edges)**.** *There is a set D of t edges of G such that exactly 2^{r-t} even subgraphs of G contain no edge of D. Every edge of D lies in some even subgraph of G.*

Proof. We choose the edges one at a time. Suppose that D has $k < r$ edges, and that the set $\mathcal{Z}_D$ of even subgraphs that avoid D has 2^{r-k} elements. This is true for $D = \emptyset$. The set $\mathcal{Z}_D$ is a subspace with more than one element. Take a nonempty $y \in \mathcal{Z}_D$ and an edge $e \in y$. The map $x \mapsto x + y$ is an involution of $\mathcal{Z}_D$ without fixed points, and exactly one of x and $x + y$ contains e. So half of the elements of $\mathcal{Z}_D$ avoid e. Add e to D. □

Lemma 3.2 (A second neighbour)**.** *Let $e = uv$ be an edge of D. Then v has a neighbour $w \neq u$ in G.*

Proof. The edge e lies in an even subgraph y. The degree of v in y is even and positive, so y has a second edge at v. Its other end is not u, because G is simple. □

Chords and triangles. For each edge e of D fix a labelling $e = uv$ of its ends and a neighbour $w_e \neq u$ of v. If u and w_e are not adjacent in G, add the edge uw_e and call it a *chord* (Figure 2). Let G^+ be the graph that we get when we add all chords, each of them once. Let a be the number of chords, so $a \leq t$. The graph G^+ is simple and connected, and it has the same vertices as G. Therefore

$$\beta(G^+) = r + a.$$

For $e \in D$ let T_e be the triangle of G^+ with the edges uv, vw_e and uw_e. Let W be the union of these triangles. Then W is a nonempty union of cycles of G^+. It has at most $3t$ edges and at most t components. It contains every edge of D and every chord. Every chord f lies on a triangle T_f of W whose other two edges are edges of G. If there are several such triangles, fix one of them.

Lemma 3.3 (Cycles through D)**.** *At most $p(G^+, W)\, 2^r$ cycles of G contain an edge of D.*

Proof. Let F be the set of chords, so $|F| = a$. For a cycle C of G that contains an edge of D, and for a set $S \subseteq F$, put

$$C_S = C + \sum_{f \in S} T_f.$$

This is an even subgraph of G^+. We show two things.

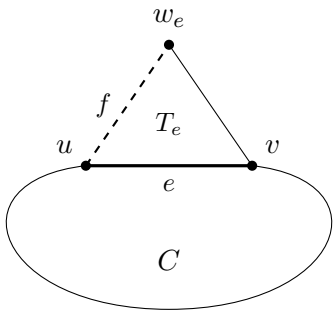


Figure 2: An edge $e = uv$ of D, a second neighbour w_e of v, and the chord $f = uw_e$ (dashed). The chord is added if u and w_e are not adjacent in G. A cycle C through e and the even subgraph $C + T_e$ have the same edges outside the triangle T_e.

The outside part of C_S is a forest. All triangles lie in W, so $C_S \setminus E(W) = C \setminus E(W)$. The cycle C contains an edge of D, and $D \subseteq E(W)$. So $C \setminus E(W)$ is a forest by Lemma 2.1(a).

The even subgraph C_S determines the pair (C, S). The cycle C contains no chord. The triangle T_f contains the chord f and no other chord. So the set of chords in C_S is S. Thus C_S determines S, and then it determines $C = C_S + \sum_{f \in S} T_f$.

Let q be the number of cycles of G that contain an edge of D. The sets C_S are $q\,2^a$ distinct even subgraphs of G^+ whose outside part is a forest. By Section 2 there are exactly $p(G^+, W)\,2^{r+a}$ such even subgraphs. Hence $q \le p(G^+, W)\,2^r$. □

Proof of Theorem 1.3. Let G be a graph of rank r, where r is large. If G is not connected, add a new vertex and join it to one vertex of each component. The new edges are bridges, so the rank and the set of cycles do not change. So we may assume that G is connected.

Let t be the least integer with $2^t > \log r$. Then $t \le \log\log r + 1$. For large r this gives

$$1 \le t \le r, \qquad (3t)^2 \le r, \qquad 100\,t \le \log r.$$

Take D as in Lemma 3.1, and take G^+ and W as above. A cycle of G avoids D or contains an edge of D. At most $2^{r-t} < 2^r/\log r$ cycles avoid D, by Lemma 3.1. At most $p(G^+, W)\,2^r$ cycles contain an edge of D, by Lemma 3.3.

The graph G^+ is connected and simple, and its rank $r^+ = r + a$ is at least r. The subgraph W is a nonempty union of cycles with at most $3t \le \sqrt{r^+}$ edges and at most $t \le (\log r^+)/100$ components. So Theorem 1.2 applies to G^+ and W, and

$$p(G^+, W) \;\le\; \frac{1}{2} + \frac{20}{\log^{(3)} r^+} \;\le\; \frac{1}{2} + \frac{20}{\log^{(3)} r}.$$

Altogether, the number of cycles of G is at most

$$\Big(\frac{1}{2} + \frac{20}{\log^{(3)} r} + \frac{1}{\log r}\Big)2^r \;\le\; \Big(\frac{1}{2} + \frac{21}{\log^{(3)} r}\Big)2^r,$$

because $\log^{(3)} r \le \log r$. Together with the lower bound, this gives $\psi(r) \sim 2^{r-1}$. □

Remark 3.4 (Why chords). One could take for W a union of cycles of G through the edges of D. Such cycles can be long, and Theorem 1.2 needs W to be small. An earlier version of this proof therefore replaced G by a cubic multigraph and searched for short cycles in it. The chords make this unnecessary: they create the short cycles, and Lemma 3.3 shows that they cost nothing.

4 The lower bound for pancyclic graphs

This section derives the lower bound of Theorem 1.1 from Theorem 1.2. The plan is as follows. A number $\Phi(R) \le 1$ measures how well a graph of rank at least R can fill an interval of cycle lengths. In a graph with many cycle lengths we find a small subgraph W that already contains the short cycles. The forest estimate for W then shows that Φ drops by a factor of almost two when the rank grows from R to 2^{256R}.

For an integer $R \ge 1$ let $\Phi(R)$ be the supremum of $(L-2)\,2^{-\beta(G)}$ over all simple graphs G with $\beta(G) \ge R$ and all integers $L \ge 2$ such that G contains a cycle of each length $3, \dots, L$. A graph of rank β has at most $2^\beta - 1$ cycles, so $\Phi(R) \le 1$. Clearly Φ is non-increasing.

Lemma 4.1 (Witness)**.** *Let $R \ge 1$ and $\varphi = \Phi(R)$, and let G contain a cycle of each length $3, \dots, L$, where $L \ge 2^{2R}$. Then G has a subgraph W that is a union of at most $2R$ cycles, with $\beta(W) \ge 2R$ and $|E(W)| \le \varphi\, 2^{2R} + 2^R + 5R$.*

Proof. Put $W_0 = \emptyset$ and $W_{i+1} = W_i \cup C_i$, where C_i is a cycle of G whose length ℓ_i is the least integer ≥ 3 that is not the length of a cycle of W_i. Stop as soon as the rank is at least $2R$. Consider a step, and let $s = \beta(W_i) < 2R$. The graph W_i contains cycles of the $\ell_i - 3$ lengths $3, \dots, \ell_i - 1$. Therefore $\ell_i - 3 \le 2^s - 1$. This gives $\ell_i \le 2^{2R-1} + 2 \le L$, so C_i exists. If $s \ge R$, then also $\ell_i - 3 \le \varphi\, 2^s$, by the definition of Φ. The cycle C_i is not a subgraph of W_i, so the rank increases at every step. So there are at most $2R$ steps, their ranks s are distinct, and

$$|E(W)| \;\le\; \sum_i \ell_i \;\le\; \sum_{s=0}^{R-1} (2^s + 2) + \sum_{s=R}^{2R-1} (\varphi\, 2^s + 3) \;\le\; \varphi\, 2^{2R} + 2^R + 5R. \qquad \square$$

Proposition 4.2 (Near-halving)**.** *For every sufficiently large integer R,*

$$\Phi(2^{256R}) \;\le\; \Big(\frac{1}{2} + \frac{20}{\log^{(2)} R}\Big)\Phi(R) + 2^{2-R}.$$

Proof. Let G be a simple graph of rank $r \ge 2^{256R}$ with a cycle of each length $3, \dots, L$. Identifying a vertex of one component with a vertex of another changes neither the rank nor the set of cycles, so we may assume that G is connected. If $L < 2^{2R}$ then $(L-2)2^{-r} < 2^{2-R}$. Otherwise let W be as in Lemma 4.1, with m edges and rank $t \ge 2R$. Then $m \le 2^{2R+1} \le \sqrt{r}$ and $\mathrm{comp}(W) \le 2R \le (\log r)/100$, so Theorem 1.2 gives $p(G, W) \le \frac{1}{2} + 20/\log^{(3)} r \le \frac{1}{2} + \delta$ with $\delta = 20/\log^{(2)} R$. Here we used $\log^{(3)} r \ge \log(8 + \log R) \ge \log^{(2)} R$. The graph G has cycles of at least $L - 2$ lengths, so Lemma 2.1(b) gives

$$\frac{L-2}{2^r} \;\le\; \frac{pm+1}{2^t} \;\le\; \Big(\frac{1}{2} + \delta\Big)(\varphi + 2^{-R} + 5R\,2^{-2R}) + 2^{-2R} \;\le\; \Big(\frac{1}{2} + \delta\Big)\varphi + 2^{2-R}. \qquad \square$$

Corollary 4.3. *There is a constant c such that $\Phi(r) \le c\, 2^{-\log^* r}$ for all $r \ge 1$.*

Proof. Fix $R_0 \ge 2^{18}$ so large that Proposition 4.2 holds for all $R \ge R_0$, and let $R_{j+1} = 2^{256R_j}$ and $\delta_j = 20/\log^{(2)} R_j$. For $j \ge 2$ we have $\log^{(2)} R_j = 8 + 256R_{j-2}$, so $\sum_j \delta_j < \infty$; also $\sum_j 2^{j-R_j} < \infty$. The numbers $a_j = 2^j \Phi(R_j)$ satisfy $a_{j+1} \le (1 + 2\delta_j)a_j + 2^{j+3-R_j}$ and are therefore bounded, say $\Phi(R_j) \le c_1 2^{-j}$.

Next, let $U_0 = R_0^2$ and $U_{j+1} = 2^{U_j}$. Then $R_j \le \sqrt{U_j}$ for all j: inductively, $R_{j+1} = 2^{256R_j} \le 2^{256\sqrt{U_j}} \le 2^{U_j/2}$ because $\sqrt{U_j} \ge 512$. Hence $\log^* R_j \le \log^* U_j = j + \log^* U_0$.

Finally let $r \ge R_0$ and let j be largest with $R_j \le r$. Then $\Phi(r) \le \Phi(R_j) \le c_1 2^{-j}$ and $\log^* r \le \log^* R_{j+1} \le j + 1 + \log^* U_0$. For $r < R_0$ use $\Phi \le 1$. $\square$

Proof of the lower bound in Theorem 1.1. Let G be a pancyclic graph with n vertices and $n+h$ edges. It is connected, so its rank is $r = h+1$, and by Corollary 4.3

$$n-2 \;\le\; \Phi(r)\,2^r \;\le\; c\,2^{\,r-\log^* r}.$$

The trivial bound $n-2 \le 2^r-1$ gives $n \le 2^{r+1} \le 2^{2^r}$, hence $\log^* n \le \log^* r + 2$. Taking logarithms, $h = r-1 \ge \log(n-2) + \log^* r - \log c - 1 \ge \log n + \log^* n - \log c - 5$. □

Formal counterpart. In the Lean code Φ is `actualRankTail`. Proposition 4.2 is `ShortProof.recurrence_of_few_component_forest_estimate`, with the same constants. Corollary 4.3 is an instance of `recurrence_logStar_bound`. Lemma 4.1 corresponds to `ShortProof.exists_budgeted_witness`, where the bookkeeping differs in detail.

5 The construction

This section proves the upper bound of Theorem 1.1. The construction follows [10]. It is a chain of beads. Bead i can be passed in two ways, and the two ways differ in length by 2^i. So k beads give 2^k consecutive path lengths for the price of k chords. A few more chords close shorter and shorter parts of the chain, and these supply the short cycles.

Proposition 5.1. *For every $n \ge 3$ there is a pancyclic graph with n vertices and at most $n + \lfloor \log n \rfloor + \log^* n + 1$ edges.*

Proof. For $3 \le n \le 6$ take a cycle $u_0u_1\cdots u_{n-1}$, with the chord u_0u_2 if $n \ge 4$ and also the chord u_0u_3 if $n = 6$.

Let $n \ge 7$, and let $k \ge 2$ be the largest integer with $2^k + k + 1 \le n$, so that $n \le 2^{k+1} + k + 1$. Let H be a cycle of length n and let $Q \subseteq H$ be a path composed of consecutive segments $Q_0, \dots, Q_{k-1}$, where Q_i has $2^i + 1$ edges and joins v_i to v_{i+1}. Thus Q has $2^k + k - 1 \le n-2$ edges. Add the chords $e_i = v_iv_{i+1}$ for $0 \le i < k$, and let N_j, the *necklace with j beads*, be the union of $Q_0, \dots, Q_{j-1}$ and $e_0, \dots, e_{j-1}$ (Figure 3).

A path from v_0 to v_j in N_j uses either Q_i or e_i for each $i < j$, and replacing Q_i by e_i shortens it by 2^i. So N_j has 2^j such paths, exactly one of each length from j to $2^j + j - 1$. Consequently a chord $f_j = v_0v_j$ with $j \ge 2$ creates cycles of all lengths in $I_j = [\,j+1,\ 2^j + j\,]$, and the path $H - Q$ together with N_k contains cycles of all lengths in $[\,n - 2^k + 1,\ n\,]$.

Let $j_0 = k$ and $j_{i+1} = \lceil \log j_i \rceil$ as long as $j_i > 2$; this sequence decreases and ends with some $j_q = 2$. Add the chords $f_{j_0}, \dots, f_{j_q}$. Since $2^{j_{i+1}} \ge j_i$, the intervals $I_{j_{i+1}}$ and I_{j_i} overlap or are adjacent, and $I_2 = [3, 6]$; so we get all lengths in $[3,\ 2^k + k]$. Together with $[\,n - 2^k + 1,\ n\,]$ this is everything, except for the single length $2^k + k + 1$ in the case $n = 2^{k+1} + k + 1$, which is supplied by one more chord, joining v_k to the neighbour of v_0 on $H - Q$.

For an integer $j \ge 3$ we have $\log^* \lceil \log j \rceil = \log^* j - 1$, so $q = \log^* k - 1$. The number of chords is therefore at most $k + \log^* k + 1 \le \lfloor \log n \rfloor + \log^* n + 1$. □

Formal counterpart. The Lean theorem `Problem1016.upper_bound_integer` states that $h(n) \le \lfloor \log n \rfloor + \log^* n + 1$ for every $n \ge 3$. Its proof formalises the construction of [10]. The proof above describes that construction in our own words. As a check on the wording, we built the graph of this section by computer for all $n \le 139$ and for eight larger values up to 522, and confirmed that it is pancyclic and has at most the stated number of edges (Appendix B).

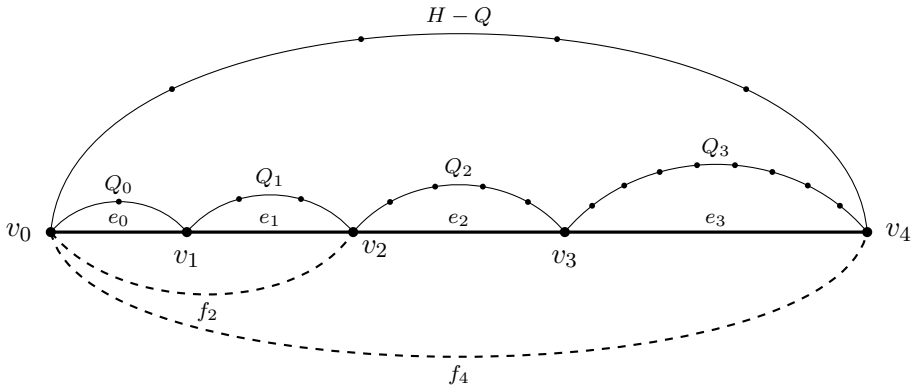


Figure 3: The construction for $n = 24$, where $k = 4$. The segments $Q_0, \dots, Q_3$ of the outer cycle H have $2, 3, 5, 9$ edges and are doubled by the chords $e_0, \dots, e_3$. The chord f_4 gives the cycle lengths $5, \dots, 20$, the rest $H - Q$ of the outer cycle gives $9, \dots, 24$, and f_2 gives $3, \dots, 6$.

6 Proof of the forest estimate

This section proves Theorem 1.2. The argument is that of [10]. It has five steps.

1. *Cuts.* For a set A of vertices, consider the event that ξ avoids every edge that leaves A. Its probability is explicit. For two disjoint sets the two events are correlated by the factor $2^{\ell-1}$, where ℓ counts the links between the sets (Lemma 6.2). One link means independence, and two links mean a factor of two.

2. *Reduction.* We pass to a graph Γ with all degrees at most 3, in which W has become a small set P of vertices (Lemma 6.5).

3. *Short cycles.* Step 1 and a second moment bound show that many disjoint short cycles in $\Gamma - P$ already force the estimate (Lemma 6.4). So we may delete a small maximal family of them, and we are left with a graph J of large girth.

4. *Supply.* The graph J has many cycles of length about $\log r$. Their total weight $\sum_C 2^{-|C|}$ is of order $\log^{(3)} r$ (Lemma 6.8). This step uses the non-backtracking matrix.

5. *Second moment.* For such a cycle C, the event that C is a component of ξ has probability about $2^{-|C|}$. If it occurs, then the part of ξ outside P is not a forest. Most pairs of these events are correlated by a factor of at most two.

The factor two in step 5 is the source of the constant $\frac{1}{2}$. Lemma 6.1 turns the bound $\mathbb{E}Z^2 \le 2\mu^2 + K\mu$ into $\mathbb{P}(Z = 0) \le \frac{1}{2} + K/(4\mu)$. Appendix A lists the Lean declarations that correspond to the steps.

In this section graphs may have loops and parallel edges. A loop adds two to the degree of its vertex and is a cycle of length one. Two parallel edges form a cycle of length two. For a set A of vertices of G we write $G[A]$ for the induced subgraph, $G - A$ for $G[V(G) \setminus A]$, and δA for the set of edges with exactly one end in A. A *region* is a nonempty set A such that $G[A]$ is connected. A region is *cyclic* if $G[A]$ contains a cycle. As before ξ is uniform on $\mathcal{Z}(G)$, and $\mathcal{O}_A$ denotes the event $\xi \cap \delta A = \emptyset$.

6.1 Two cuts share at most one bit

Lemma 6.1. *Let $Z \geq 0$ be a random variable with $\mu = \mathbb{E}Z > 0$ and $\mathbb{E}Z^2 \leq 2\mu^2 + K\mu$. Then $\mathbb{P}(Z = 0) \leq \frac{1}{2} + K/(4\mu)$.*

Proof. By the Cauchy–Schwarz inequality $\mu^2 \leq \mathbb{E}Z^2 \cdot \mathbb{P}(Z > 0)$, so $\mathbb{P}(Z = 0) \leq 1 - \mu/(2\mu + K) \leq \frac{1}{2} + K/(4\mu)$. □

For disjoint regions A, B of a connected graph G let $\ell(A, B)$ be the number of *links* between them: the edges with one end in A and the other in B, and the components of $G - (A \cup B)$ that are joined by an edge to A and by an edge to B.

Lemma 6.2 (Link formula)**.** *Let A and B be disjoint regions of a connected graph G. Then $\mathbb{P}(\mathcal{O}_A) = 2^{\mathrm{comp}(G-A)-|\delta A|} \geq 2^{-|\delta A|}$ and*

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\ell(A,B)-1}\,\mathbb{P}(\mathcal{O}_A)\,\mathbb{P}(\mathcal{O}_B).$$

Conditionally on $\mathcal{O}_A$, the set $\xi \cap E(G[A])$ is uniform on $\mathcal{Z}(G[A])$; conditionally on $\mathcal{O}_A \cap \mathcal{O}_B$, the sets $\xi \cap E(G[A])$ and $\xi \cap E(G[B])$ are independent and uniform on $\mathcal{Z}(G[A])$ and $\mathcal{Z}(G[B])$.

Proof. The even subgraphs that avoid δA are the sums of an even subgraph of $G[A]$ and an even subgraph of $G - A$. So $\mathbb{P}(\mathcal{O}_A) = 2^{\beta(G[A])+\beta(G-A)-\beta(G)}$, and counting edges and vertices shows that the exponent is $\mathrm{comp}(G - A) - |\delta A|$. In the same way

$$\mathbb{P}(\mathcal{O}_A \cap \mathcal{O}_B) = 2^{\beta(G[A])+\beta(G[B])+\beta(G-A-B)-\beta(G)} = 2^{\mathrm{comp}(G-A-B)+1-|\delta A|-|\delta B|+e(A,B)},$$

where $e(A, B)$ is the number of edges between A and B. Let a, b and k be the numbers of components of $G-A-B$ that are joined to A only, to B only, and to both. Then $\mathrm{comp}(G-A-B) = a+b+k$, while $\mathrm{comp}(G-A) = a+1$ and $\mathrm{comp}(G-B) = b+1$, and the formula follows since $\ell(A, B) = k+e(A, B)$. The conditional laws are read off from the two direct sum decompositions. □

So two cut events are independent when there is one link, correlated by a factor of exactly two when there are two, and more strongly correlated only for pairs with at least three links. The next lemma says that such pairs are rare.

Lemma 6.3 (Exceptional partners)**.** *Let A be a region of a connected graph G, and let $B_1, \ldots, B_k$ be pairwise disjoint regions, disjoint from A, with $\ell(A, B_i) \geq 3$ for all i. Then $k \leq |\delta A|$.*

Proof. Contract A to a vertex a and each B_i to a vertex b_i, and delete the loops that arise. This changes neither $d = |\delta A|$ nor the numbers $\ell(A, B_i)$, because the sets B_j with $j \neq i$ are connected and are contracted inside components of $G - (A \cup B_i)$. Let F be a spanning forest of $G - a$, and for every edge $e = au$ at a attach a new leaf τ_e to u. Call the resulting forest F^+ and its d new leaves *terminals*.

Fix i. Every link between a and b_i yields a path of F^+ from b_i to a terminal. For an edge $e = ab_i$ it is the edge $b_i\tau_e$. For a component D of $G - \{a, b_i\}$ joined to both, choose an edge $e = au$ with $u \in D$ and take the path of F^+ from b_i to τ_e; after its first vertex it stays in $D \cup \{\tau_e\}$. Paths that come from different links therefore start with different edges at b_i, and the union of any two of them is a path between two terminals. Let $F^\circ \subseteq F^+$ be the union of all paths between terminals. As $\ell(a, b_i) \geq 3$, the vertex b_i has degree at least 3 in F°. All leaves of F° are terminals, and a forest with at most d leaves has at most $d - 2$ vertices of degree at least 3. Hence $k \leq d - 2$. □

Lemma 6.4 (Many small cuts)**.** *Let $A_1, \dots, A_k$ be pairwise disjoint cyclic regions of a connected graph G with $|\delta A_i| \le d$ for all i. Then*

$$\mathbb{P}(\xi \cap E(G[A_i]) \text{ is a forest for every } i) \;\le\; \frac{1}{2} + \frac{(d+1)\,2^{d-1}}{k}.$$

Proof. Let E_i be the event that $\mathcal{O}_{A_i}$ holds and $\xi \cap E(G[A_i]) \neq \emptyset$. On E_i the set $\xi \cap E(G[A_i])$ is a nonempty even subgraph, hence not a forest, so it suffices to bound $\mathbb{P}(Z = 0)$ for $Z = \sum_i \mathbf{1}_{E_i}$. By Lemma 6.2, $p_i := \mathbb{P}(E_i) = \mathbb{P}(\mathcal{O}_{A_i})\big(1 - 2^{-\beta(G[A_i])}\big) \ge 2^{-d-1}$ and $\mathbb{P}(E_i \cap E_j) = 2^{\ell(A_i,A_j)-1} p_i p_j$ for $i \neq j$. If $\ell(A_i, A_j) \le 2$ this is at most $2p_ip_j$. Otherwise we use $\mathbb{P}(E_i \cap E_j) \le p_i$; by Lemma 6.3 this happens for at most d indices j, for each i. Hence $\mathbb{E}Z^2 \le \mu + 2\mu^2 + d\mu$ with $\mu = \mathbb{E}Z \ge k\,2^{-d-1}$, and Lemma 6.1 gives the claim. □

6.2 Reduction to a subcubic graph

Lemma 6.5. *Let G and W be as in Theorem 1.2, with $m = |E(W)|$ and $c = \mathrm{comp}(W)$. There are a connected multigraph Γ with $\beta(\Gamma) = r$ and all degrees at most 3, and a set $P \subseteq V(\Gamma)$, such that*

(i) *every vertex outside P has degree 3, $|P| \le 2m$, and $\Gamma[P]$ has at most c components;*

(ii) *$p(G, W) \le \mathbb{P}\big(\xi_\Gamma \cap E(\Gamma - P)$ is a forest$\big)$, where ξ_Γ is uniform on $\mathcal{Z}(\Gamma)$.*

In particular $2r - 2 \le |V(\Gamma)| \le 2r + 2m$.

Proof. Splitting. Replace every vertex v of degree $d \ge 4$ by a path $v^1v^2 \cdots v^{d-2}$ and distribute the d edges at v along it in some order: two to v^1, one to each of $v^2, \dots, v^{d-3}$, and two to v^{d-2}. If $v \in V(W)$, let the edges of W at v come first in this order, so that they end in an initial segment $v^1, \dots, v^i$ of the path with $i < \deg_W(v)$. Let G' be the new graph; its degrees are at most 3. Let P consist of these initial segments and of the vertices of W that were not split. Then $|P| \le \sum_v \deg_W(v) = 2m$, every edge of W joins two vertices of P, and $G'[P]$ has at most c components.

Contracting the new path edges maps $\mathcal{Z}(G')$ bijectively onto $\mathcal{Z}(G)$, so we may take $\xi_G = \xi_{G'} \cap E(G)$. Every edge of $G' - P$ is a new path edge or an edge of $E(G) \setminus E(W)$. If $\xi_G \setminus E(W)$ is a forest, then so is its union with the set of all new path edges, because contracting these edges, which form a forest, gives back $\xi_G \setminus E(W)$. Hence $p(G, W) \le \mathbb{P}(\xi_{G'} \cap E(G' - P)$ is a forest$)$.

Cleaning. As long as possible, delete a vertex of degree one outside P, or replace a vertex of degree two outside P and its two edges by a single edge. An even subgraph avoids a pendant edge and contains both or none of the two edges at a vertex of degree two, so these operations induce bijections of the cycle spaces. Let Γ be the final multigraph. Each edge of $\Gamma - P$ is a path of G' all of whose vertices lie outside P; so if ξ_Γ contains a cycle of $\Gamma - P$, then the corresponding $\xi_{G'}$ contains a cycle of $G' - P$. This proves (ii).

Edges with both ends in P are never removed, so the vertices of P keep degree at least two, and the number of components of the graph induced on P does not increase. The vertices outside P now have degree exactly 3. Finally $3|V(\Gamma)| - |P| \le 2|E(\Gamma)| \le 3|V(\Gamma)|$ and $|E(\Gamma)| = |V(\Gamma)| + r - 1$ give the bounds on $|V(\Gamma)|$. □

6.3 Parameters, and removing the short cycles

From now on Γ and P are as in Lemma 6.5, $\xi = \xi_\Gamma$, and r is large. All $o(\cdot)$ and $O(\cdot)$ refer to $r \to \infty$ and are uniform in G and W. Put

$$x = \log r, \qquad y = \log x, \qquad z = \log y,$$
$$D = \lfloor x/10 \rfloor, \qquad L = \lfloor x\sqrt{y} \rfloor, \qquad s = \lfloor (D-1)/2 \rfloor, \qquad \varepsilon = 20/z.$$

Recall that $|P| \le 2\sqrt{r}$ and $c \le x/100$. Let $\mathcal{F}$ be the event that $\xi \cap E(\Gamma - P)$ is a forest. We assume that

$$\mathbb{P}(\mathcal{F}) > \tfrac{1}{2} + \varepsilon \tag{2}$$

and derive a contradiction; by Lemma 6.5(ii) this proves Theorem 1.2.

Let $\mathcal{Q}$ be a maximal family of pairwise disjoint cycles of $\Gamma - P$ of length at most D, loops and pairs of parallel edges included, and let S be the union of their vertex sets. For $Q \in \mathcal{Q}$ the set $V(Q)$ is a cyclic region of Γ with $|\delta V(Q)| \le |Q| \le D$, because its vertices have degree 3. On $\mathcal{F}$ the set $\xi \cap E(\Gamma[V(Q)])$ is a forest for every Q, so Lemma 6.4 and (2) give $|\mathcal{Q}| < (D+1)2^{D-1}/\varepsilon \le \sqrt{r}$. Hence $|S| \le D\sqrt{r}$.

Let $J = \Gamma - (P \cup S)$. By the maximality of $\mathcal{Q}$, J is a simple graph of girth greater than D. It has $v = 2r + O(\sqrt{r}\log r)$ vertices, each of degree 3 in Γ, so its *deficit* satisfies

$$d_J := \sum_{u \in V(J)} (3 - \deg_J u) \;\le\; 3(|P| + |S|) = O(\sqrt{r}\log r). \tag{3}$$

6.4 Cycles of moderate length

A walk $u_0 u_1 \cdots u_\ell$ is *non-backtracking* if $u_{i+1} \ne u_{i-1}$ for $0 < i < \ell$. Let $N_\ell(u, w)$ be the number of non-backtracking walks of length ℓ from u to w in J.

Lemma 6.6. *If $\ell > s$, then $N_\ell(u,w) \le 3 \cdot 2^{\ell-s-1}$ for all u, w. Consequently, for every vertex u of J, the cycles C of J through u with $|C| \le L$ satisfy $\sum_C 2^{-|C|} \le L\, 2^{-s}$.*

Proof. There are at most $3 \cdot 2^{\ell-s-1}$ ways to choose the first $\ell - s$ steps. The last s steps form a non-backtracking walk of length s between two given vertices. Such a walk is a path, because s is less than the girth, and there is at most one such path, because two of them would give a cycle of length at most $2s < D$. For the second statement, a cycle of length ℓ through u can be traversed from u in two directions, so there are at most $\frac{1}{2}N_\ell(u,u)$ of them, and $\ell > D > s$. □

Let B be the non-backtracking matrix of J. Its rows and columns are indexed by the directed edges $(i \to j)$ of J, two for each edge, and the entry in row $(i \to j)$ and column $(j' \to k)$ is 1 if $j' = j$ and $k \ne i$, and 0 otherwise.

Lemma 6.7. *For every even $\ell \ge 2$,*

$$\operatorname{tr} B^\ell \;\ge\; \bigl(2 - d_J/v\bigr)^\ell - 3v\, 2^{\ell/2}.$$

Proof. Let A be the adjacency matrix of J and Δ the diagonal matrix of its degrees, and put $M(\lambda) = \lambda^2 I - \lambda A + \Delta - I$.

(1) *If $M(\lambda)g = 0$ for a real $\lambda > 1$ and a vector $g \ne 0$, then λ is an eigenvalue of B.* A direct computation shows that $f(i \to j) = \lambda g(j) - g(i)$ satisfies $Bf = \lambda f$; and $f = 0$ would give $g(i) = \lambda g(j) = \lambda^2 g(i)$ along every edge, hence $g = 0$ (at an isolated vertex, $M(\lambda)g = 0$ already says $(\lambda^2 - 1)g = 0$).

(2) *If $Bf = \lambda f$ with $f \ne 0$ and $\lambda^2 \ne 1$, then $M(\lambda)g = 0$ for $g(j) = \sum_{k \sim j} f(j \to k)$, and $g \ne 0$.* The eigenvalue equation at $(i \to j)$ and at $(j \to i)$ reads $\lambda f(i \to j) = g(j) - f(j \to i)$ and $\lambda f(j \to i) = g(i) - f(i \to j)$. Eliminating $f(j \to i)$ gives $(\lambda^2 - 1)f(i \to j) = \lambda g(j) - g(i)$, and summing this over the neighbours j of i gives $M(\lambda)g = 0$.

(3) *B has a real eigenvalue $\lambda_1 \ge 2 - d_J/v$.* The matrix $M(2) = 2(\Delta - A) + (3I - \Delta)$ is positive semidefinite, while $\mathbf{1}^\top M(\lambda)\mathbf{1} = (\lambda - 1)(v\lambda - 2v + d_J)$ vanishes at $\lambda^* = 2 - d_J/v > 1$. The smallest

eigenvalue of the symmetric matrix $M(\lambda)$ is continuous in λ, at most 0 at λ^* and at least 0 at 2. So $M(\lambda_1)$ is singular for some $\lambda_1 \in [\lambda^*, 2]$, and (1) applies.

(4) *Every non-real eigenvalue λ of B satisfies $|\lambda| \le \sqrt{2}$.* With g as in (2), $g^*M(\lambda)g = 0$ is a quadratic equation $a\lambda^2 - b\lambda + c' = 0$ with real coefficients $a = g^*g > 0$, $b = g^*Ag$ and $c' = g^*(\Delta - I)g \le 2a$. Its roots are λ and $\bar{\lambda}$, so $|\lambda|^2 = c'/a \le 2$.

Now $\operatorname{tr} B^\ell$ is the sum of the ℓth powers of the $2|E(J)| \le 3v$ eigenvalues of B. For even ℓ the real eigenvalues contribute nonnegative terms, one of which is at least $(2 - d_J/v)^\ell$, and the non-real ones contribute at least $-3v\,2^{\ell/2}$ in total. □

Facts (2) and (4) are part of the Ihara–Bass theory [3, 13]; the use of the test vector $\mathbf{1}$ in (3), which makes the argument work without any assumption of regularity or expansion, is from [10].

Lemma 6.8 (Cycle weight)**.** *Let $\mathcal{C}$ be the set of cycles of J of length at most L. Then*

$$V := \sum_{C \in \mathcal{C}} 2^{-|C|} \;\ge\; (\tfrac{1}{8} - o(1)) \ln y.$$

Proof. $\operatorname{tr} B^\ell$ is the number of closed walks $u_0u_1 \cdots u_\ell = u_0$ in J with $u_{i+1} \ne u_{i-1}$ for all i modulo ℓ. A cycle of length ℓ gives 2ℓ of them; let n_ℓ be the number of the others. Then

$$T := \sum_{\ell \le L} \frac{\operatorname{tr} B^\ell}{2\ell\, 2^\ell} = V + \sum_{\ell \le L} \frac{n_\ell}{2\ell\, 2^\ell}.$$

Consider a walk counted by n_ℓ, and let j be the least index with $u_j \in \{u_0, \dots, u_{j-1}\}$, say $u_j = u_i$. Then $j < \ell$, the walk $Q = u_iu_{i+1} \cdots u_j$ is a cycle, of length q say, and $u_ju_{j+1} \cdots u_\ell u_1 \cdots u_i$ is a non-backtracking closed walk of length $\ell - q > 0$ from u_i to itself. It contains a cycle, so $\ell - q > D > s$. The original walk is determined by Q, by the vertex u_i and a direction on Q ($2q$ choices), by the complementary walk (at most $3 \cdot 2^{\ell-q-s-1}$ choices, by Lemma 6.6), and by the position of u_0 on the latter (at most ℓ choices). Hence

$$\sum_{\ell \le L} \frac{n_\ell}{2\ell\, 2^\ell} \;\le\; \sum_{\ell \le L} \sum_{Q \in \mathcal{C}} \tfrac{3}{2}\, |Q|\, 2^{-|Q|-s} \;\le\; \tfrac{3}{2}\, L^2\, 2^{-s}\, V = o(V),$$

because $2^{-s} \le 8\, r^{-1/20}$ and $L^2 \le x^2 y$.

On the other hand, for even ℓ with $6x \le \ell \le L$, Lemma 6.7 and (3) give

$$2^{-\ell} \operatorname{tr} B^\ell \;\ge\; \Big(1 - \frac{d_J}{2v}\Big)^\ell - 3v\, 2^{-\ell/2} \;\ge\; 1 - \frac{L\, d_J}{2v} - 3v\, r^{-3} = 1 - o(1).$$

All terms of T are nonnegative, so $T \ge (1 - o(1)) \sum 1/(2\ell)$ over the even ℓ in $[6x, L]$, and this sum is $(\frac{1}{4} - o(1)) \ln(L/6x) = (\frac{1}{8} - o(1)) \ln y$. Therefore $V = (1 - o(1))T \ge (\frac{1}{8} - o(1)) \ln y$. □

6.5 The second moment

For $C \in \mathcal{C}$ let I_C be the event that C is a component of ξ. As the vertices of C have degree 3 in Γ, this is the event $E(C) \subseteq \xi$; it is also the event that $\mathcal{O}_{V(C)}$ holds and $\xi \cap E(\Gamma[V(C)]) = E(C)$. Put

$$a_C = 2^{-|C|}, \qquad p_C = \mathbb{P}(I_C), \qquad X_C = \frac{a_C}{p_C}\, \mathbf{1}_{I_C}, \qquad Z = \sum_{C \in \mathcal{C}} X_C.$$

The event I_C is given by $|C|$ linear conditions on $\xi \in \mathcal{Z}(\Gamma)$, which are satisfied by $\xi = C$; so $p_C \ge a_C$. Hence $0 \le X_C \le 1$, $\mathbb{E}X_C = a_C$ and $\mu := \mathbb{E}Z = V$. On $\mathcal{F}$ no cycle of $J \subseteq \Gamma - P$ is contained in ξ, so $Z = 0$, and by (2)

$$\mathbb{P}(Z = 0) > \tfrac{1}{2} + \varepsilon. \tag{4}$$

Let $C \neq C'$ be in $\mathcal{C}$. If they share a vertex, then $I_C \cap I_{C'} = \emptyset$. If they are disjoint, Lemma 6.2, applied to the regions $V(C)$ and $V(C')$ of Γ, gives $\mathbb{P}(I_C \cap I_{C'}) = 2^{\ell(C,C')-1} p_C p_{C'}$ and therefore

$$\mathbb{E}(X_C X_{C'}) = a_C a_{C'}\, 2^{\ell(C,C')-1}, \tag{5}$$

where $\ell(C, C')$ is the number of links between $V(C)$ and $V(C')$ in Γ.

Lemma 6.9 (Few links)**.** *$\ell(C, C') \leq c + o(x)$ for all disjoint $C, C' \in \mathcal{C}$.*

Proof. Let $U = V(C) \cup V(C')$. Every vertex of U lies on exactly one edge outside its own cycle, so at most $2L$ edges leave U. At most c components of $\Gamma - U$ meet P, because $P \cap U = \emptyset$ and so each component of $\Gamma[P]$ lies in one of them. Let Y run over the other components and let d_Y be the number of edges between Y and U; then $\sum_Y d_Y \leq 2L$. Put $t_0 = \lfloor y/2 \rfloor$.

Large cuts. At most $2L/t_0$ components have $d_Y > t_0$.

Small cuts, cyclic. The components Y with $d_Y \leq t_0$ that are cyclic are pairwise disjoint cyclic regions inside $\Gamma - P$ with $|\delta Y| \leq t_0$. On $\mathcal{F}$ each $\xi \cap E(\Gamma[Y])$ is a forest, so by Lemma 6.4 and (2) there are fewer than $(t_0 + 1)2^{t_0-1}/\varepsilon$ of them.

Small cuts, trees. Let Y induce a tree, with $d_Y \leq t_0$. No $Q \in \mathcal{Q}$ meets Y, since Q is a connected subgraph of $\Gamma - U$ and would lie in Y. Hence $Y \subseteq V(J)$. As all degrees are 3, we have $3|Y| = 2(|Y| - 1) + d_Y$, that is, $|Y| = d_Y - 2$. So if Y is joined to both cycles, there is a path R from C to C' through Y with at most t_0 edges. For a link that is an edge, let R be that edge. The paths R obtained in this way lie in J and are pairwise disjoint. Divide each of C and C' into at most $8L/D$ blocks of at most $D/4$ consecutive vertices. If two of the paths had their ends in the same block of C and in the same block of C', they would form, together with two arcs inside these blocks, a cycle of J of length less than $2t_0 + D/2 \leq D$. So there are at most $(8L/D)^2$ such paths.

Altogether

$$\ell(C, C') \;\leq\; c + \frac{2L}{t_0} + \frac{(t_0+1)2^{t_0-1}}{\varepsilon} + \Big(\frac{8L}{D}\Big)^2 = c + O\Big(\frac{x}{\sqrt{y}}\Big) + O(yz\sqrt{x}) + O(y) = c + o(x). \qquad \square$$

Proof of Theorem 1.2. Call a disjoint pair $C, C' \in \mathcal{C}$ *exceptional* if $\ell(C, C') \geq 3$. Fix C, and let $\mathcal{E}$ be a maximal family of pairwise disjoint exceptional partners of C. By Lemma 6.3, $|\mathcal{E}| \leq |\delta V(C)| \leq L$, so the union U_C of their vertex sets has at most L^2 vertices, and every exceptional partner of C meets U_C. By Lemma 6.6, the sum of $a_{C'}$ over all $C' \in \mathcal{C}$ that meet U_C is at most $L^3 2^{-s}$. With (5) and Lemma 6.9,

$$\sum_{C' \text{ exceptional}} \mathbb{E}(X_C X_{C'}) \;\leq\; a_C\, 2^{c+o(x)} L^3\, 2^{-s} = a_C\, 2^{\,c - x/20 + o(x)} = o(a_C),$$

because $c \leq x/100$. For the other disjoint pairs $\mathbb{E}(X_C X_{C'}) \leq 2a_C a_{C'}$, intersecting pairs contribute nothing, and $\mathbb{E}X_C^2 \leq \mathbb{E}X_C = a_C$. Summing over C we get $\mathbb{E}Z^2 \leq 2\mu^2 + (1 + o(1))\mu$. By Lemmas 6.1 and 6.8,

$$\mathbb{P}(Z = 0) \;\leq\; \frac{1}{2} + \frac{1 + o(1)}{4\mu} \;\leq\; \frac{1}{2} + \frac{2 + o(1)}{\ln y} \;<\; \frac{1}{2} + \frac{20}{z}$$

for large r, since $\ln y = z \ln 2$. This contradicts (4). $\square$

Remark 6.10. The assumption $|E(W)| \leq \sqrt{r}$ entered only through $|P| \leq 2\sqrt{r}$ in (3), where all that is needed is $L\, d_J = o(v)$; the proof therefore works unchanged for $|E(W)| \leq r/\log^2 r$. Similarly the constant 20 can be replaced by any constant larger than $2/\ln 2$. We have kept the statement of [10], which is the one that has been formalised.

7 Exact and nearly exact counting

By Lemma 2.1 the cycles of G have at most $2^{r-t}(pm + 1 - p)$ lengths. Theorem 1.2 says that $p \le \frac{1}{2} + o(1)$ when r is large. This section studies the other end of the scale, where p can be 1. It proves Theorem 1.4: part (a) is Proposition 7.1, and part (b) is Proposition 7.2. Nothing in this section is formalised.

7.1 Two terminals

Proposition 7.1. *Let N be a connected graph and let $s \neq t$ be two of its vertices. Then N has at most $2^{\beta(N)}$ paths from s to t. If it has exactly $2^{\beta(N)}$, then $N - E(A)$ is a forest for every path A from s to t, and every such path has at least $\beta(N)$ edges.*

Proof. A path from s to t is an $\{s,t\}$-join, and the joins form a coset of $\mathcal{Z}(N)$, so there are $2^{\beta(N)}$ of them. Suppose that all joins are paths, and let A be one of them. If $N - E(A)$ contained a cycle C, then $A \cup C = A + C$ would be a join that is not a path. So $N - E(A)$ is a forest, and since deleting an edge lowers the rank by at most one, $\beta(N) \le |A|$. □

Equivalently, an edge f of a graph G lies on at most $2^{\beta(G)-1}$ cycles, and if it lies on that many, then each of them has at least $\beta(G)$ edges. Equality in Proposition 7.1 holds for the necklace N_j of Section 5 between v_0 and v_j, for K_4 minus an edge between its two vertices of degree two, and for $K_{3,3}$ minus an edge between the ends of the deleted edge.

This explains the shape of the construction. The necklace N_k turns k units of rank into 2^k path lengths, the most that k units can give. By Proposition 7.1, no gadget that does this has a path with fewer than k edges between its terminals. So the cycles through N_k have length greater than k, and the lengths $3, \dots, k$ must be produced elsewhere. The construction uses a necklace with about $\log k$ beads for them. Its own short lengths require a third necklace, and so on.

Each of these necklaces costs a factor of two. Consider a necklace N_j that is closed by a chord f_j, together with f_j and the chords of the smaller necklaces. The rest of the graph is attached to this part at the two vertices v_0 and v_j only. So half of the outside parts cannot be used (Example 2.2). These $\log^* k$ halvings are the term $\log^* n$ of Theorem 1.1.

7.2 No waste

Call an outside part *wasted* if it is not the outside part of any cycle. The bound of $m\,2^{r-t}$ cycle lengths can be exact only if no outside part is wasted. This cannot go on for long.

Proposition 7.2. *Let G be a connected graph and let $W \subseteq G$ be a subgraph that contains a cycle. Suppose that no outside part is wasted. Then every nonempty even subgraph of $G - E(W)$ is a cycle, and*

$$\beta(G) - \beta(W) \;\le\; |V(W)| + 3.$$

Proof. Let z be a nonempty even subgraph of $G - E(W)$. Then z is its own outside part. By assumption some cycle C has outside part z, so $z \subseteq C$. A cycle that contains a nonempty even subgraph is equal to it. So z is a cycle.

In particular, two edge-disjoint cycles of $G - E(W)$ would have a union that is an even subgraph but not a cycle. So the graph $G' = G - E(W)$, on the vertex set $V(G)$, has no two edge-disjoint cycles. By Lemma 7.3 below, $|E(G')| \le |V(G)| + 3$. Now $\beta(G) = |E(G)| - |V(G)| + 1$ and $\beta(W) \ge |E(W)| - |V(W)| + 1$. Hence $\beta(G) - \beta(W) \le |E(G')| - |V(G)| + |V(W)| \le |V(W)| + 3$. □

Lemma 7.3 (Erdős and Pósa [7])**.** *A multigraph with n vertices and at least $n+4$ edges contains two edge-disjoint cycles.*

Proof. We include the short proof. Let H be a counterexample with $|V(H)|+|E(H)|$ minimal, and let n be its number of vertices. Then H has exactly $n+4$ edges and no vertex of degree at most one. It has no vertex of degree two either: such a vertex could be replaced, together with its two edges, by a single edge, or it carries a loop, and the rest of H contains a second cycle. So $2(n+4) \geq 3n$, that is, $n \leq 8$. Let C be a shortest cycle of H. Since $H - E(C)$ is a forest, $n+4-|C| \leq n-1$, so $|C| \geq 5$. Thus H is a simple graph of minimum degree at least 3 and girth at least 5, and such a graph has at least $1+3+6=10$ vertices, a contradiction. □

The first conclusion of Proposition 7.2 is the graphic case of the fact that a binary code in which every nonzero word is minimal is intersecting [1, §2]. The second is sharp up to the additive constant. If W is a cycle $w_1 \cdots w_m$ and G consists of W and a path of length two from w_i to w_{i+1} for every i modulo m, then no outside part is wasted and $\beta(G)-\beta(W) = |V(W)|$. If G is K_5 minus an edge and W is the triangle on its three vertices of degree four, then no outside part is wasted and $\beta(G) - \beta(W) = |V(W)|+1$. An exhaustive search found nothing better for a triangle W in graphs with at most eight vertices.

So the trivial bound of $m\,2^{r-t}$ cycle lengths can be exact for at most $m+3$ ranks beyond W, while by Theorem 1.2 it is off by a factor of almost two as soon as $r \geq m^2$, for large m and W as in that theorem.

7.3 A defect budget

The last statement of this section is a stability version. If a graph comes close to the trivial bound, then almost every step of a greedy sequence looks like a bead of the necklace.

Let G contain a cycle of each length $3, \ldots, L$, and let $\Theta = (L-2)\,2^{-\beta(G)}$. A *greedy sequence* is a sequence of subgraphs $W_1 \subset W_2 \subset \cdots \subset W_Q$ of G in which W_1 is a triangle, $W_{q+1} = W_q \cup C_q$ for a cycle C_q of G whose length ℓ_q is the least integer ≥ 3 that is not the length of a cycle of W_q, and W_Q is the first term that contains cycles of all lengths $3, \ldots, L$. Write $m_q = |E(W_q)|$ and $t_q = \beta(W_q)$, and let

$$\rho_q = \frac{m_q+2}{2^{t_q}}, \qquad \kappa_q = 1 - \frac{m_{q+1}-m_q}{m_q+2}, \qquad j_q = t_{q+1} - t_q - 1.$$

Proposition 7.4 (Defect budget)**.** *For every greedy sequence:*

(i) $0 < \kappa_q < 1$, $j_q \geq 0$ *and* $\rho_{q+1} = (1-\frac{1}{2}\kappa_q)\,2^{-j_q}\rho_q$ *for* $1 \leq q < Q$*;*

(ii) $\Theta < \rho_q$ *for all* q*, and therefore* $\sum_{q=a}^{Q-1}(\frac{1}{2}\kappa_q + j_q \ln 2) < \ln(\rho_a/\Theta)$ *for every* a*;*

(iii) *if* $j_q = 0$ *and* $\kappa_q \leq \eta$*, then* C_q *has at least* $(1-\eta)(m_q+2)$ *edges outside* W_q *and fewer than* $\eta(m_q+2)$ *edges in* W_q*, and* W_q *has cycles of at least* $(1-\eta)(m_q+2)-3$ *different lengths.*

Proof. (i) Let $P_q = E(C_q) \setminus E(W_q)$, so that $|P_q| = m_{q+1} - m_q \geq 1$. The graph W_q has cycles of all lengths $3, \ldots, \ell_q - 1$ and no cycle longer than m_q. Hence $|P_q| \leq \ell_q \leq m_q + 1$ and $0 < \kappa_q < 1$. Also $j_q \geq 0$, because $C_q \in \mathcal{Z}(W_{q+1}) \setminus \mathcal{Z}(W_q)$, and the identity restates $m_{q+1}+2 = (2-\kappa_q)(m_q+2)$.

(ii) By Lemma 2.1(b) with $W = W_q$ and $p \leq 1$, the cycles of G have at most $m_q 2^{\beta(G)-t_q}$ lengths, so $\Theta \leq m_q 2^{-t_q} < \rho_q$. By (i), $\rho_Q \leq \rho_a \exp\bigl(-\sum_{q=a}^{Q-1}(\frac{1}{2}\kappa_q + j_q \ln 2)\bigr)$.

(iii) Here $|P_q| = (1-\kappa_q)(m_q+2)$, the number of edges of C_q in W_q is $\ell_q - |P_q| \leq m_q + 1 - |P_q|$, and W_q has cycles of at least $\ell_q - 3 \geq |P_q| - 3$ lengths. □

Fix $\eta \in (0,1)$, and call a step *perfect* if $j_q = 0$ and $\kappa_q \le \eta$. Suppose that $\Theta \ge \rho_a/K$ for some a and some $K > 1$. An imperfect step contributes more than $\eta/2$ to the sum in (ii). So all but $(2/\eta)\ln K$ of the steps after W_a are perfect. By (iii), a perfect step adds at least $(1-\eta)(m_q+2)$ new edges, in a cycle that has fewer than $\eta(m_q+2)$ edges in W_q. At that moment W_q has cycles of at least $(1-\eta)(m_q+2)-3$ lengths. If W_q is connected, then $j_q = 0$ also means that the new edges form one path or cycle that meets W_q in at most two vertices. These are the properties of a bead of the necklace.

8 Problems

Error terms. The error terms in Theorems 1.2 and 1.3 are of order $1/\log^{(3)} r$. The truth may be much smaller. By Example 2.2 the left side of (1) can be as large as $\frac{1}{2} + 2^{1-r}$, and Horak and Kim [9] proved that $\psi(r) \ge 2^{r-1} + c\, r\, 2^{r/2}$. We know no examples with larger error terms.

The weak error term has a definite cause in the proof. The gain μ comes from cycles with between $6\log r$ and $\log r\sqrt{\log\log r}$ edges. Each length ℓ in this range contributes about $1/(2\ell)$ to μ. Lemma 6.9 dictates the upper end of the range. So μ is only of order $\log^{(3)} r$.

Problem 8.1. Is the error term in (1) at most r^{-c} for some $c > 0$? Is $\psi(r) \le 2^{r-1} + 2^{(1-c)r}$?

Between exactness and the forest estimate. Proposition 7.2 and Theorem 1.2 describe the same quantity at two scales. No outside part is wasted for at most $|V(W)| + 3$ ranks beyond W. About half of the outside parts fail to be forests once $r \ge |E(W)|^2$, or already once $r \ge |E(W)|\log^2 r$ by Remark 6.10. We know no proof of the factor $\frac{1}{2}$ that avoids the second moment computation of Section 6, except in special cases such as Example 2.2. A stability version of Proposition 7.1 would be a step in that direction.

Problem 8.2. Let $\varepsilon > 0$. If a connected graph N has at least $(\frac{1}{2} + \varepsilon)2^{\beta(N)}$ paths between two vertices s and t, must each of these paths have at least $c(\varepsilon)\beta(N)$ edges?

Problem 8.3. Can the constant 3 in Proposition 7.2 be replaced by 1?

A The formal statements

The Lean 4 code is in the repository [12]. This paper refers to commit `756d5cc` on the branch `feat/cycle-count-bound`. The code uses Lean v4.19.0 [14] and a pinned revision of Mathlib [15]. It has 256 modules and 42,495 lines. The nine modules in `Erdos1016/CycleCount/` are new in this commit; they have 1,554 lines and contain the proof of Theorem 1.3.

A.1 The statements behind Theorem 1.3

The following text is copied from `Erdos1016/CycleCount/Statement.lean`, `Asymptotic.lean` and `Erdos1016/Main.lean`, without the comments and the proofs.

```
def cycleEdgeSets {V : Type*} (G : SimpleGraph V) : Set (Set (Sym2 V)) :=
  {s | ∃ (v : V) (p : G.Walk v v), p.IsCycle ∧ s = {a | a ∈ p.edges}}

def numCycles {V : Type*} (G : SimpleGraph V) : ℕ := (cycleEdgeSets G).ncard

def CycleCountStatement : Prop :=
```

```
  ∃ r₁ : ℕ, ∀ (n : ℕ) (G : SimpleGraph (Fin n)) (r : ℕ),
    r + n = Nat.card G.edgeSet + Nat.card G.ConnectedComponent → r₁ ≤ r →
    (numCycles G : ℝ) ≤
      (1 / 2 + 21 / Real.logb 2 (Real.logb 2 (Real.logb 2 (r : ℝ)))) * 2 ^ r

def cycleCounts (r : ℕ) : Set ℕ :=
  {c | ∃ (n : ℕ) (G : SimpleGraph (Fin n)),
    r + n = Nat.card G.edgeSet + Nat.card G.ConnectedComponent ∧ numCycles G = c}

def maxCycles (r : ℕ) : ℕ := sSup (cycleCounts r)

def AsymptoticCycleCount : Prop :=
  Tendsto (fun r : ℕ => (maxCycles r : ℝ) / 2 ^ r) atTop (𝓝 (1 / 2))

theorem cycleCountTheorem : CycleCount.CycleCountStatement
theorem cycleCountAsymptotic : CycleCount.AsymptoticCycleCount
```

Four remarks help to compare this with Theorem 1.3.

- `SimpleGraph`, `Walk` and `IsCycle` are Mathlib's. A walk with `IsCycle` is a closed walk of length at least 3 that repeats no vertex except its start. The set `cycleEdgeSets G` contains the edge set of each such walk once. So `numCycles G` is the number of cycles in the sense of this paper, and not the number of closed walks.
- The hypothesis `r + n = ...` says that $r = \beta(G)$. It avoids subtraction of natural numbers.
- The threshold is existentially quantified, as r_1 is in the text. In the proof it is $\max(r_0, 2^{2048})$, where r_0 is the threshold of the forest estimate.
- In Lean, `sSup` of an empty or unbounded set of natural numbers is 0. This does not make `maxCycles` degenerate. The code proves that `cycleCounts r` is bounded by 2^r (`cycleCounts_bddAbove`) and that its supremum is attained for $r \geq 3$ (`maxCycles_mem`).

The theorem `cycleCountTheorem_general` states the same bound for simple graphs on every finite vertex type.

A.2 The forest estimate

Theorem 1.2 is the theorem `ShortProof.fewComponentForestEstimate`, which has the following type.

```
def FewComponentForestEstimate : Prop :=
  ∃ rmin : ℕ, ∀ (G : PhysicalGraph), G.IsConnected → rmin ≤ G.cycleRank →
    ∀ F : Finset G.CycleWord, F.Nonempty →
      (witnessSupportEdges G F).card ^ 2 ≤ G.cycleRank →
      100 * (Nat.card (cycleUnionGraph G F).ConnectedComponent : ℝ) ≤
        Real.logb 2 (G.cycleRank : ℝ) →
      outsideForestProbability G (witnessSupportEdges G F) ≤
        (1 / 2 : ℝ) + 20 / tripleLog G.cycleRank
```

Here `PhysicalGraph` is the project's type of finite simple graphs with numbered vertices and edges. The family `F` is a nonempty finite set of cycles of `G`, and W is their union. The set `witnessSupportEdges G F` is $E(W)$. The graph `cycleUnionGraph G F` is W on the vertices that

the cycles meet. The number `outsideForestProbability G E` is the number of even subgraphs x such that $x \setminus E$ is a forest, divided by $2^{\beta(G)}$. Finally `tripleLog r` is $\log^{(3)} r$. The statement of Theorem 1.1 in Lean is described in [10, §1.2].

A.3 Repeating the check

The repository contains a verifier. It rebuilds every module from source. It compares the types of the endpoint theorems with statements that are written out a second time in `audits/TheoremAudit.lean`. It prints the axioms of 59 declarations, among them all endpoints in Table 1, and it accepts only `propext`, `Classical.choice` and `Quot.sound`. It rejects proofs that contain `sorry`. The commands are:

```
git clone https://github.com/JWKNT/erdos1016.git && cd erdos1016
git checkout 756d5ccd35c3f916e9e6ae0c00bb7992fc10d6fa
elan toolchain install leanprover/lean4:v4.19.0
lake exe cache get
python3 -m unittest discover -s scripts -p 'test_*.py'
python3 scripts/verify.py --fresh --jobs 2
```

On 10 November 2026 this check passed for the commit above on a machine of the authors, with a clean working tree. [The commit has not been pushed yet. Push it, and cite the public CI run here, before this paper is circulated.]

A.4 Steps of the proofs

Table 2 names the Lean declaration for each step. Namespaces are omitted, and files are relative to `Erdos1016/`. The rows for Section 3 match the text closely, with three differences. The formal proof uses $r \le \beta(G^+) \le r + a$ in place of the equality. It states Lemma 3.3 together with the count of the cycles that avoid D. It writes t as $\lfloor \log \lfloor \log r \rfloor \rfloor + 1$, which is the same number.

The rows for Sections 4 to 6 are taken from [10, Appendix A]. We checked that these declarations exist at the commit above. We did not compare each of them with our wording. In these parts the formal proof replaces O-terms by explicit constants, and intermediate estimates differ in detail from the text. The statements of Theorems 1.1 and 1.2, of Proposition 4.2 and of Corollary 4.3 agree with the Lean statements.

In this paper	Declaration	File
Lemma 3.1	`exists_independent_edges`	`CycleCount/IndependentEdges`
Lemma 3.2	`exists_apex`	`CycleCount/IndependentEdges`
Chords and triangles	`addChords`, `cycleRank_addChords_le`, `exists_triangle_family`	`CycleCount/ChordExtension`, `CycleCount/TriangleFamily`
Lemma 3.3	`card_cycleWord_le`	`CycleCount/ConnectedBound`
Theorem 1.3, parameters	`witness_parameters`, `cycleCountBound_of_forestEstimate`	`CycleCount/Bound`
Theorem 1.3, Mathlib form	`cycleCountStatement_of_bound`	`CycleCount/Statement`
Lower bound $\psi(r) \geq 2^{r-1}$	`exists_graph_many_cycles`	`CycleCount/LowerBound`
$\psi(r) \sim 2^{r-1}$	`asymptoticCycleCount_of_statement`	`CycleCount/Asymptotic`
Lemma 4.1	`exists_budgeted_witness`	`Extremal/Witness/CapacityBudget`
Proposition 4.2	`recurrence_of_few_component_forest_estimate`	`Extremal/Recurrence/WitnessReduction`
Corollary 4.3	`recurrence_logStar_bound`	`Extremal/Recurrence/LinearExponentialDecay`
Proposition 5.1	`upper_bound_integer`	`Extremal/UpperBound`
Lemma 6.2	`multigraph_zeroCut_pair`	`CycleSpace/Graphical/MultigraphCutCorrelation`
Lemma 6.3	`exceptional_partner_card_le`	`CycleSpace/Graphical/DisjointRegionCuts`
Lemma 6.4	`smallCut_forest_bound`	`Probability/Moments/SmallCutRegionBound`
Lemma 6.5	`exists_marked_subcubic_reduction`	`Graph/Cubicization/MarkedReduction`
Lemma 6.7	`normalized_even_trace_lower`	`Nonbacktracking/Spectrum/DeficitTrace`
Lemma 6.8	`cycleWordMass_ge_log_ratio`	`Cycles/Counting/LowDegreeTraceCycles`
Lemma 6.9	`actual_cycle_links_le`	`Cycles/Geometry/ActualCycleLinkBound`
Theorem 1.2	`second_moment_le`, `forest_le_half_add_twenty`, `fewComponentForestEstimate`	`Probability/Moments/SeedFamilyAvoidance`, `Probability/Avoidance/FewComponentForest`

Table 2: Steps of the proofs and the corresponding Lean declarations.

B Tests on small graphs

We tested the statements that are not formalised, and our wording of some that are, on small graphs. The scripts are in the directory `paper-checks` that accompanies this manuscript. They use exact arithmetic, except for the two scripts on the non-backtracking matrix, which compute eigenvalues in floating point. All tests passed. A passed test is evidence, not a proof.

- *Section 3* (`check_chords.py`). 300 random connected graphs with 4 to 8 vertices and every

admissible t, which gives 834 cases. The script builds D, the chords, G^+ and W. It checks Lemmas 3.1, 3.2 and 3.3, and the stated properties of G^+ and W.

- *Lemma 2.1 and Proposition 7.2* (`check_counting.py`). 4000 random connected graphs with 4 to 8 vertices, each with a random subgraph W. The script checks the number of outside parts, the bound of Lemma 2.1(b), and the bound $2^r p + 2^{r-t}(1-p)$ on the number of all cycles, which follows from the proof of that lemma. In the 2027 cases without a wasted outside part it checks both conclusions of Proposition 7.2.

- *Proposition 7.2, exhaustively* (`nowaste_exh.py`, `nowaste_exh2.py`). For connected W we have $\beta(G) - \beta(W) - |V(W)| = |E(G) \setminus E(W)| - |V(G)|$. The scripts enumerate connected graphs G that contain a fixed triangle W. They cover all such graphs with 5 or 6 vertices. For 7 vertices they cover the graphs with at least 8 edges outside W, and for 8 vertices those with at least 10. For a fixed four-cycle W they cover all graphs with 6 vertices, and the graphs with 7 vertices and at least 8 edges outside W. Graphs with more than $|V(G)| + 3$ edges outside W are skipped, because Lemma 7.3 excludes them. Without a wasted outside part, the largest value of $\beta(G) - \beta(W) - |V(W)|$ was 1 for the triangle and 0 for the four-cycle.

- *Proposition 5.1* (`check_construction.py`). The graph of Section 5 for all $n \le 139$ and for $n \in \{263, 264, 265, 300, 500, 520, 521, 522\}$. Each graph is pancyclic, and the number of chords never exceeds $\lfloor \log n \rfloor + \log^* n + 1$.

- *Proposition 7.4* (`check_defect.py`). Greedy sequences in the graph of Section 5 for six values of n up to 140, and in 1442 random connected graphs with 5 to 9 vertices. The script checks part (i), and part (ii) for $a = 1$.

- *Lemmas 6.2 and 6.3* (`check_links.py`). 400 random connected multigraphs with 5 to 9 vertices and 5221 pairs of disjoint regions with at most three vertices each. The script checks the link formula and the bound on exceptional partners.

- *Lemmas 6.6 and 6.7* (`check_nb.py`, `check_nb2.py`). 292 random graphs of maximum degree 3 with 10 to 40 vertices, and the Petersen, Heawood and Tutte–Coxeter graphs. The scripts check the walk bound, the two statements on eigenvalues, and the trace inequality for even $\ell \le 24$.

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